MILITARY SPECIFICATION
FILTER UNIT, GAS-PARTICULATE, EMD, 600 CFM, M9A2

1. SCOPE

1.1 This specification covers a filter unit for protective shelters.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel Strapping, Flat.
UU-P-271	-	Paper, Wrapping, Waterproofed Kraft.
PPP-B-636	-	Box, Fiberboard.
PPP-T-60	-	Tape: Pressure-Sensitive, Adhesive, Waterproof, For Packaging.

MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Flexible.
MIL-B-131	-	Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-E-16298	-	Electric Machines Having Rotating Parts; Packaging of.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

EDGEWOOD ARSENAL

DL5-19-1858 - Filter Unit, Gas-Particulate, EMD, 600 CFM, M9A2.
(All drawings referenced in this specification are listed on this drawing.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM STANDARDS

E84-61 - Surface Burning Characteristics of Building Materials.

(Copies of this standard may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-19-1858 and on subsidiary drawings shall conform to the specifications and the specific characteristics set forth on the drawings and in this specification.

3.1.1.1 Plywood. The plywood used in the construction of the inlet and outlet plenum assemblies shall have a flame spread classification of 25 or less when tested as specified in ASTM Standard E84.

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3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing DL5-19-1858 and on subsidiary drawings.

3.2 Manufacture. The filter unit shall be manufactured and assembled as specified in Drawing D5-19-1858.

3.3 Filter unit performance. The filter unit shall produce 600 ± 30 cubic feet per minute (cfm) of air against a static head of not less than 1.5 inches of water when tested as specified in 4.5.

3.4 Preproduction. When specified in the contract or order, a preproduction sample shall be produced for examination and test. The preproduction sample shall be manufactured using the same materials, methods, and equipment as will be used during regular production (see 6.2).

3.5 Workmanship. The filter unit shall be free from damage such as bent, strained, or warped metal or wooden sections and foreign matter such as dirt, grease, oily or viscous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. A lot shall consist of the filter units produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of M14A1 gas-particulate filters shall be represented in any one lot of finished filter units.

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4.4 Inspection provisions.

4.4.1 Examination and nondestructive tests. Each filter unit shall be examined in accordance with the list of defects and tested in accordance with 4.5. Any item found defective shall be rejected.

4.4.2 List of defects.

4.4.2.1 Filter unit, gas-particulate, EMD, 600 CFM, M9A2 (Dwg D5-19-1858).

1. Component missing or incorrect
2. Component incorrectly assembled
3. Protective coating missing
4. Workmanship (3.5)
5. Marking missing
6. Vaneaxial fan-motor mounting holes incorrectly located (Dwg C5-19-1859, Dwg C5-19-1860 as applicable)
7. Hardware inoperative
8. Gasket material frayed or broken

4.5 Filter unit performance test. The filter unit shall be tested for the specified airflow delivery against the required static head using Edgewood Arsenal Q60 Collective Protector Airflow Tester.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Filter unit assembly (less hose assembly and technical publications). The vaneaxial fan and motor of the filter unit assembly shall be cleaned and dried in accordance with MIL-E-16298. Both ends of the filter unit assembly shall be covered with barrier material conforming to class 1 of MIL-B-131. The material shall be secured in place with pressure sensitive tape conforming to class 1 of PPP-T-60.

5.1.1.2 Hose assembly and technical publications. Technical publications shall be preserved and packaged in accordance with method IC-3 of MIL-P-116. The hose assembly as shown on Drawing C5-19-1841 shall have each end covered with barrier material conforming to type I, grade A, class 2 of MIL-B-121 and

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secured in place with tape conforming to PPP-T-60. The hose assembly shall be preserved and packaged in accordance with method IC-5 of MIL-P-116 except that the fiberboard container shall enclose both the technical publications and hose assembly. The container shall be closed and sealed in accordance with the appendix of PPP-B-636.

5.1.2 Level C. The hose assembly, technical publications, and filter unit assembly shall be packaged to provide adequate protection against corrosion, deterioration, and physical damage from the supply source to the first activity for immediate use.

5.2 Packing.

5.2.1 Level A. The filter unit assembly as shown on Drawing D5-19-1858, with the hose assembly and technical publications packaged as specified in 5.1.1.1 and 5.1.1.2 shall be packed within a crate conforming to type II, class 1, style a of MIL-C-104. A waterproof interior shroud conforming to class E2 of UU-P-271 shall be placed over the filter unit assembly. The shroud shall be secured to the filter unit assembly with tape conforming to PPP-T-60. The filter unit assembly shall be secured to the wood base of the crate with eight 3/8-inch diameter lag or carriage bolts. The hose assembly container with the technical publications contained therein shall also be secured to the wood base of the crate with metal strapping conforming to type I, class B, grade 2 of QQ-S-781.

5.2.2 Level C. The filter unit assembly as shown on Drawing D5-19-1858 with the hose assembly and technical publications packaged as specified in 5.1.2 shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers a filter unit for removing toxic gas and aerosols from atmospheric air supplied to protective shelters.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Whether a preproduction sample is required.

(1) Quantity required.
(2) Time allowed for suppliers submission of sample for Government test and evaluation after award of contract.
(3) Name and address of test facilities and shipping instructions when testing is performed by the Government.
(4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Component specifications. The following is a list of the component specifications for the M9A2 filter units:

Title	Number
Filters, Gas, (60, 120, and 250 CFM Modules) C10, C31 and C21	MIL-F-51213
Filters, Gas, 600, 1200, 2500, and 5000 CFM; C22R1, C32R1, C29R1 and C23R1	MIL-F-51214
Filters, Particulate, 600, 1200, 2500, and 5000 CFM; C18R1, C19R1, C30R1 and C20R1	MIL-F-51215
Filters, Gas-Particulate, 600, 1200, 2500, and 5000 CFM; M14A1, M15A1, M16A1 and M17A1	MIL-F-51224

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No 4240-A205

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Commanding Officer
US Army Edgewood Arsenal
ATTN: SMUEA-QAWS
Edgewood Arsenal, Maryland 21010

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